

Matrix inequalities and norms

Sharp additive triangle constant for an odd number of contractions

Difficulty: challenging **Importance:** interesting to specialist

Status: [Lean verified](#)

Rating rationale: Sharpness for every odd summand count needs a new extremal construction or obstruction; the payoff is concentrated in operator triangle inequalities.

Resolution — 2026-09-11

Affirmative result by Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. **Independent proof review:** **PASS**.

The sharp additive contraction constant is $c_k = k/4$ for every $k \geq 2$, including every odd summand count. Exact dimension-two extremizers match the universal upper bound.

The exact target is resolved. The original statement and source evidence are retained below; its former difficulty rating is historical.

Primary manuscript: [complete proof PDF](#), [standalone TeX](#), Theorem 1.1 and its proof; [authorship and scope](#). The [independent review](#) checks the full original argument and records its hash. The draft was AI-assisted. That original review was informal; the later Lean verification is documented below. External human peer review is not claimed. [Submission record](#).

Lean proof and verification evidence - 2026-09-12

The complete original odd-summand conjecture is **Lean verified**. The [proof at revision 901ba5f](#) proves that $k/4$ is the least member of the full admissible-constant set for every $k \geq 2$, then identifies its actual real infimum. This includes every odd $k \geq 3$, with all positive dimensions and every tuple of complex contractions. Genuine principal CFC moduli and Euclidean operator norms are used; no literature inequality is assumed as an unproved premise.

Lean formalization: George Stepaniants, Department of Computing and Mathematical Sciences, California Institute of Technology, Pasadena, California, USA, with AI-agent assistance. **Matthew J. Colbrook** retains authorship of the mathematical proof and result; **Jean-Christophe Bourin and Eun-Young Lee** retain the original conjecture and prior-bound credit.

The eight [checked exports](#), each with prefix `NLA.MI03.`, are:

- `modulus_semantics`: genuine positive square root, square and norm identities.
- `contraction_modulus`: positive modulus defects for every complex contraction.
- `positive_decomposition`: the exact universal Gram-sum and positive-square identity.
- `universal_upper_bound`: admissibility of $k/4$ for every $k \geq 2$.
- `root_of_unity_data`: exact complex roots and their vanishing geometric sums.
- `sharpness_witness`: true moduli, unit operator norms and full sums of the two-dimensional extremizers.
- `sharp_constant`: actual least admissible constant and infimum, for every $k \geq 2$.
- `odd_contraction_conjecture`: the complete original assertion.

Two independent agents approved the [statements and completed proof](#). [Linux run 34722618003](#) matched all eight declarations with the sandboxed Comparator and replayed the solution in Lean's default kernel. The [original artifacts and operational audit](#) bind all 173 input files and both actual isolation/rejection-control suites. All 16 [internal/public transitive axiom reports](#) use exactly `propext`, `Classical.choice` and `Quot.sound`. The operational reviewer also served as final proof referee 1; these roles do not count as a third mathematical referee. External human peer review is not claimed.

The pins are [Lean 4.33.1](#), [LeanCert 621a43d](#) and [Mathlib 0df444a](#). LeanCert audits this exact proof's kernel trust; there is **no numerical interval certificate**. Symbolic identities and all- k roots avoid numerical phase approximation or interval subdivision. The source's extra three-dimensional Hermitian extremizers and rank classification are outside these exports; the complete original target is covered. See the [project guide](#), [manifest](#) and [dependency pins](#). From the immutable verified revision on a documented non-root Linux host, reproduce with:

```
tools/lean/bootstrap.sh /absolute/path/to/nla-lean-tools
tools/lean/selftest.sh /absolute/path/to/nla-lean-tools
tools/lean/verify.sh \
  matrix-inequalities-and-norms/MI-03/lean \
  /absolute/path/to/nla-lean-tools
```

Problem statement

For each integer $k \geq 2$, let c_k be the infimum of all $c \geq 0$ such that

$$\left| \sum_{j=1}^k A_j \right| \leq cI_n + \sum_{j=1}^k |A_j|$$

for every $n \geq 1$ and all $A_1, \dots, A_k \in \mathbb{C}^{n \times n}$ satisfying $\|A_j\|_2 \leq 1$. Here $|A| = (A^*A)^{1/2}$, $\|\cdot\|_2$ denotes the operator norm, and $X \leq Y$ means $Y - X$ is positive semidefinite.

Is $c_k = k/4$ for every odd integer $k \geq 3$?

Why it matters

The scalar absolute-value triangle inequality fails in matrix order. The optimal additive correction quantifies that failure for uniformly bounded summands, in a form usable in positive block-matrix estimates.

References

1. J.-C. Bourin and E.-Y. Lee, *Diagonal and off-diagonal blocks of positive definite partitioned matrices*, arXiv:2307.02034v3 (15 December 2023), Corollary 4.4 and the conjecture immediately following Remark 4.5. [Primary text](#).
2. E.-Y. Lee, *How to compare the absolute values of operator sums and the sums of absolute values?*, *Operators and Matrices* 6(3) (2012), 613–619; §2 supplies related triangle inequalities. [Primary paper](#), [DOI](#).

Status check — 2026-09-10

Corollary 4.4 establishes $c_k \leq k/4$ for every k and sharpness for even k ; the source separately conjectures sharpness for all odd $k > 1$. Its latest version remains v3. Searches included *Bourin Lee contractions odd k constant $k/4$, three contractions $3/4$ sharp conjecture*, and *Bourin contractions sharp 2025 2026*. The 2026 symmetric-modulus papers concern different inequalities. No resolution of this additive odd-summand problem was located.

Audit update (2026-09-10): Rechecked the conjecture after Remark 4.5 in Bourin–Lee and searched for odd-summand sharpness results. The proved even-summand statement is outside the target, which already restricts to odd k . This is a bounded literature check, not a proof that no solution exists.